PRD: Teacher Attendance App

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Overview and goals

The Teacher Attendance App is a web app and installable PWA where a teacher registers students and their guardians and records attendance for two breaks each school day, replacing the paper monthly register.

Problem: the paper register is slow to total and easy to lose, and nothing should ever block the teacher from marking it.

Goals for v1

- A teacher marks a class of 40 for one break in under 1 minute.
- A teacher imports a full class and its guardians from Excel or CSV in one step.
- The app runs fully online, so a teacher just needs an internet connection to mark attendance.
- Every record shows its date, break and time in the Africa/Mogadishu time zone.
- A Monthly Register view and export match the paper sheet teachers use today.
- Nothing is ever blocked: no windows, no locks, no stuck sessions.

Users and scope

The only user is the teacher. Each teacher signs in with email and password and sees only their own classes, students and attendance. There is no school entity, no admin role and no tenant model. It is one app on one database, and every record carries its owner (`teacher_id`).

In scope for v1

- Login, dashboard and settings, with Resend-powered password reset email
- Classes, students and guardians (manual entry and Excel/CSV import)
- Take attendance per class, per school day, per break, with any date
- Three statuses: Present, Absent and Excused (Fasax) with an optional note
- A PDF receipt generated from a template after each break is saved, downloadable and emailable via Resend
- Monthly Register view and export that mirrors the paper sheet
- Attendance records, reports and export to Excel and PDF
- Installable PWA (online only, no offline mode)

- Somali and English interface

Out of scope for v1

- Schools, admins, multi-tenancy
- Subjects and per-subject attendance
- Attendance windows, cutoff times and locking
- Parent or student portal, SMS or WhatsApp alerts
- Grading, payments, QR or biometric check-in

Core rules

These rules shape every screen: school days, two breaks, three statuses, exact timestamps, and one time zone.

Rule	Requirement
School days	Saturday to Wednesday by default. Thursday and Friday are off. Marking on an off day shows a warning but is never blocked.
Two breaks	Each class has two attendance records per school day. Break 1 runs 1:00 to 3:00 PM and Break 2 runs 3:00 to 6:00 PM. The times are editable and only choose the default break. Nothing locks.
No subjects	Attendance is per class, per day and per break. There are no subjects.
Statuses	Present, Absent and Excused (Fasax). Every student starts as Present.
Excused	The student told the teacher in advance and has permission to be off. It applies to one break or the whole day, with an optional note.
Percentage	Attendance % = Present ÷ (Present + Absent). Excused counts as neither Present nor Absent, so it is left out of both sides. Reports show the Excused count separately.
Flexible	The teacher can take or edit attendance for any day and break at any time, today or in the past.
Recorded at	The system stamps the save time automatically. The teacher cannot change it.

Rule	Requirement
Edited at	Every edit adds its own timestamp. A quiet history is kept and never blocks the teacher.
Time zone	All times display in Africa/Mogadishu (UTC+3, no daylight saving). Stored as `timestamptz` in UTC.
Duplicates	One record per class, date and break. A second attempt opens the existing one to edit.

Functional requirements

The app has a Login screen and ten sidebar items. Each one is specified below in order.

1. Login

- Email and password, with "Forgot password" and a Somali/English toggle.
- A successful login redirects to the Dashboard. A failed login shows a clear error.

2. Dashboard

- **KPI cards:** total students, total classes, breaks today (taken vs not taken), today's attendance rate, this month's attendance rate, Excused today, and students at risk (below the threshold, default 75%).
- **Charts:** 30-day attendance trend (line), Present/Absent/Excused share (donut), Break 1 vs Break 2 rate (bar), rate by class (bar).
- **Lists:** today's breaks with a "Not taken" badge and a Take button, low-attendance students, and recent activity with Africa/Mogadishu times.
- A date-range filter at the top applies to every KPI and chart.

3. Take Attendance

- The teacher picks a class, a date (default: today) and a break (default: the one that matches the current time, Break 1 from 1:00 PM and Break 2 from 3:00 PM).
- Every student starts as Present. One tap cycles Present, Absent and Excused. An optional note can be added per student.
- Marking Excused offers "this break only" or "whole day".
- Shortcuts: mark all present, mark all absent, search, and live counters.
- The teacher must be online to save. The full flow, including the PDF generated after each save, is in the next section.

4. Monthly Register

- A grid that mirrors the paper sheet: one row per student (number, name, guardian phone) and, for each school day, two columns, Break 1 and Break 2.
- Each cell shows P, A or E (Joogid, Maqan, Fasax) with a symbol as well as a color. An empty cell means "not taken".
- The header shows the class, the month, and the Wiilal / Gabdho / Wadar counts (boys, girls, total).
- Tap any cell to mark or correct it.
- Export to PDF or Excel, laid out like the paper register.

5. Classes

- List with search. Create, edit and archive. Fields: class name (for example F3C) and optional description.
- Class detail shows its students, boy, girl and total counts, and attendance rate.

6. Students

- Fields: name, student number, gender, class, and guardian.
- Table with search, class filter and attendance percentage. Add, edit and archive.
- Student profile shows the full attendance history, Excused days, and a call-guardian button.

7. Guardians

- Guardians have their own table, linked to students. Fields: name and up to two phone numbers.
- One guardian can be linked to several students, for example two brothers who share one number.
- List with search and the linked students. Add, edit and archive.
- When adding a student, the teacher picks an existing guardian or creates a new one.

8. Attendance Records

- Lists every break with its date, break number, recorded-at time, Present, Absent and Excused counts, and an edited flag.
- Filters: class, date range, break, and "Not taken". Open any record to view or edit it.

9. Reports

- Daily, per class, per student, Excused, and low-attendance reports.

- Date-range filter, attendance percentages (Excused left out), and export to Excel and PDF.

10. Import

- Download the template (student name, student number, gender, guardian name, guardian phone 1 and 2), upload an Excel or CSV file, review a preview with errors and duplicates, then confirm.
- Keep an import history with a downloadable error file.

11. Settings

- Profile, change password, language (Somali or English), low-attendance threshold (default 75%), and the time zone shown as Africa/Mogadishu.
- Logout sits at the bottom of the sidebar.

Take Attendance flow

A teacher can reach the screen three ways: from the sidebar, from "Take" on a break in today's Dashboard list (class pre-filled), or by opening a past record in Attendance Records to edit it.

```
flowchart TD
    A[Choose class] --> B[Set date and break]
    B --> C{Record already exists?}
    C -- Yes --> D[Open existing record to edit]
    C -- No --> E[Load students, all Present]
    D --> F[Mark Absent or Excused]
    E --> F
    F --> G[Save]
    G --> H[Saved, recorded-at stamped]
    H --> I[PDF receipt generated]
```

The teacher's path from start to save takes seven steps.

1. Choose a class.
2. Set the date and the break. The date defaults to today and the break to the one that matches the current time. Both can be changed, including to a past date.
3. If a record already exists for that class, date and break, the app asks "Already taken. Open to edit?"
4. The student list loads with every student marked Present, and a counter shows Present, Absent and Excused.

5. Tap a student to cycle Present, Absent, Excused. Excused offers "this break only" or "whole day", and a note icon adds an optional note. Search finds a student fast.
6. "Mark all present" and "Mark all absent" reset the list.
7. Save shows a summary of the date, break and counts, then a confirmation "Saved at 2:05 PM", with options to take the next break or go to the Dashboard.
8. A PDF receipt for that break is generated from a fixed template (class, date, break, and each student's status) and offered as a download; the teacher can also email it to themselves via Resend.

Edit path. Opening a past break loads its saved statuses, date and break number. The teacher changes anything and saves. The recorded-at time stays, an edited-at time is added, and nothing is locked.

Edge cases

Situation	Behavior
Class has no students	Show "No students yet" with buttons to add or import.
Unsaved changes on leave	Ask "Discard changes?"
No internet connection	Show "You're offline. Reconnect to save attendance." and disable Save until the connection returns.
Save fails	Show a clear error and let the teacher retry without losing what they entered.
Student added after a break was taken	Show them as "Not marked" when the record is edited.
Same break edited on two devices	The later save wins, and both edits stay in the history.
Thursday or Friday	Show "Not a school day. Continue?" and allow it.
Student Excused for the whole day	Break 2 opens with that student already marked Excused.

Data model

Eight entities cover v1. Every table except Teacher carries `teacher_id`, and every query filters on it.

Entity	Key fields
Teacher	id, name, email, password hash, language, school days, break times, password_reset_token, password_reset_expires
Class	id, teacher_id, name, description, archived
Student	id, teacher_id, class_id, guardian_id, name, student_no, gender, archived
Guardian	id, teacher_id, name, phone_1, phone_2, archived
Attendance Session	id, teacher_id, class_id, session_date, break_no (1 or 2), recorded_at, edited_at, pdf_generated_at
Attendance Record	id, session_id, student_id, status (present, absent, excused), excused_scope (break or day), note
Import Batch	id, teacher_id, file name, row counts, error file, created_at
Audit Log	id, teacher_id, entity, entity_id, action, old and new values, changed_at

All timestamps are `timestamptz`, stored in UTC and shown in Africa/Mogadishu. Each session is unique on class, session_date and break_no.

Demo seed data (10 students)

#	Student	No.	Parent	Parent phone
1	Abdirahman Mohamed Ali	STU-001	Mohamed Ali Nur	+252 61 555 0101
2	Hodan Abdullahi Nur	STU-002	Amina Hassan	+252 61 555 0102
3	Ayaan Hassan Warsame	STU-003	Hassan Warsame	+252 62 555 0103
4	Mohamud Ahmed Jama	STU-004	Ahmed Jama	+252 61 555 0104
5	Fadumo Yusuf Omar	STU-005	Yusuf Omar	+252 65 555 0105
6	Abdullahi Ibrahim Farah	STU-006	Halima Farah	+252 61 555 0106
7	Sagal Osman Duale	STU-007	Osman Duale	+252 62 555 0107
8	Ismail Abdi Hersi	STU-008	Abdi Hersi	+252 61 555 0108
9	Khadra Ali Mohamud	STU-009	Ali Mohamud	+252 66 555 0109

#	Student	No.	Parent	Parent phone
10	Yasin Farah Abdi	STU-010	Farah Abdi	+252 61 555 0110

Non-functional requirements

The app must stay fast and usable on low-end phones with weak connectivity.

Area	Requirement
PWA	Installable from the browser with a home-screen icon and web manifest. Runs full screen. Online only: the app requires an internet connection to load and save.
Performance	Attendance screen loads in under 3 seconds on a 3G connection. A class of 40 saves in under 2 seconds.
Capacity	Up to 5,000 students and 500,000 attendance records per teacher without slowdown.
Security	HTTPS only, hashed passwords, session expiry, and every query scoped by `teacher_id`. Daily database backups.
Localization	Somali and English, switchable at any time. Somali phone number format (+252).
Time	All times shown in Africa/Mogadishu. Stored in UTC. Correct even if the phone's own time zone is wrong.
Devices	Works on current Chrome and Safari, phone first, and scales to tablet and desktop.
Accessibility	Large tap targets, readable contrast, and no color-only status cues.

Tech stack and architecture

The app is one Next.js application on one PostgreSQL database, deployed once.

Layer	Choice
Front end	Next.js (App Router), responsive and mobile first
PWA	Web manifest for install; no offline queue, no service-worker caching of data
API	Next.js route handlers or server actions
Database	PostgreSQL, with migrations
Auth	Email and password, secure session cookies
Email	Resend API for transactional email: password reset links and the attendance PDF receipt
Import and export	Excel/CSV parsing on import, Excel and PDF generation for reports
Hosting	One deployment with HTTPS and daily database backups

```
flowchart LR
    A[Teacher phone or browser] --> B[Next.js PWA]
    B --> D[Next.js API]
    D --> E[(PostgreSQL)]
```

The teacher's device talks directly to the app over the internet; there is no local queue or sync step.

Success metrics, milestones and open questions

v1 succeeds if a teacher can take attendance quickly online, without ever being blocked by the app's own rules.

Success metrics

- Take attendance for a class of 40 in under 1 minute.
- Import a full class in one step, with errors clearly reported.
- Attendance screen loads in under 3 seconds on 3G.

Milestones

Phase	Scope
1. Design	Wireframes for Login, Dashboard and Take Attendance, and the database schema
2. MVP	Login, classes, students, guardians, manual entry, Take Attendance with two breaks and Excused
3. Data in	Excel/CSV import with preview and error file
4. Insight	Dashboard KPIs and charts, Monthly Register, Attendance Records, Reports and export
5. PWA	Install as a PWA, Somali and English
6. Pilot	Real teachers use it, feedback, fixes, release

Open questions

- [ ] Can a student belong to more than one class, or only one?
- [ ] Should students start as Present, or start blank until the teacher marks them?
- [ ] Confirm: one guardian can have several students and up to two phone numbers.
- [ ] Confirm: Late is removed, Break 1 ends at 3:00 PM, and Gender is added for the Wiilal / Gabdho / Wadar counts.
- [ ] Do we keep the quiet edit history in v1, or drop it for simplicity?
- [ ] Should guardian alerts (SMS or WhatsApp) come in a later version?